

Competency list.

How physiology works, and what keeps you steady

Name _____

Date _____

These are the twelve things you have to be able to do by Sunday. Every exam question, lab and case in Week 1 comes from this list. Tick a box only when it is true.

HOW TO USE THE THREE BOXES

Read. You have read it, in the book or the notes, and it made sense at the time.

Drew it. You have drawn or worked it with the notes open beside you.

From memory. You did it on blank paper with nothing open. This is the only box that predicts an exam score, and it is the one to chase. A row with the first two ticked and the third empty is exactly where your study time should go.

Foundations of Physiology HOMEOSTASIS AND CONTROL

8 competencies

1	Physiology and levels of function Define physiology and place a given process at the correct level of organization from molecule to organism, then state the level at which that process is best explained.	☐ READ	☐ DREW IT	☐ FROM MEMORY
2	Structure and function relationship Predict how a change in the structure of a molecule, cell, tissue, or organ alters its function, using examples from two different organ systems.	☐ READ	☐ DREW IT	☐ FROM MEMORY
3	Homeostasis defined Define homeostasis, regulated variable, and setpoint, and distinguish homeostasis from chemical equilibrium and from steady state.	☐ READ	☐ DREW IT	☐ FROM MEMORY
4	Feedback loop components Diagram a negative feedback loop labeling stimulus, sensor, afferent path, integrating center, efferent path, effector, and response, and trace body temperature or blood glucose through every step.	☐ READ	☐ DREW IT	☐ FROM MEMORY
5	Negative and positive feedback Distinguish negative from positive feedback by the direction of the response and give a physiological example of each, explaining why a positive feedback loop needs an outside event to end it.	☐ READ	☐ DREW IT	☐ FROM MEMORY
6	Feedforward control and acclimatization Explain anticipatory feedforward control and acclimatization and identify which one is operating in a given scenario.	☐ READ	☐ DREW IT	☐ FROM MEMORY
7	Local and reflex control pathways Distinguish a local control pathway from a long distance reflex pathway and classify a given response as one or the other.	☐ READ	☐ DREW IT	☐ FROM MEMORY

8 Mass balance

Apply the mass balance equation to a solute or to body water and calculate the intake, production, and output combination that holds the amount in the body constant.

☐
READ

☐
DREW
IT

☐
FROM
MEMORY

Quantitative Skills for Physiology METHOD AND MEASUREMENT

4 competencies

9 Units and unit conversion

Convert among the units used in physiology including molarity, osmolarity, milliequivalents, mmHg, liters per minute, and percent solutions.

☐
READ

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DREW
IT

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FROM
MEMORY
10 Graphing and data interpretation

Construct a labeled graph with the independent variable on the x axis, and read slope, direction, and trend from a physiological data set.

☐
READ

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DREW
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FROM
MEMORY
11 Experimental design and controls

Identify the hypothesis, independent variable, dependent variable, and control condition in a physiology experiment and state what the control rules out.

☐
READ

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DREW
IT

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FROM
MEMORY
12 Measurement error and variability

Distinguish random from systematic error and explain why physiological measurements are repeated and averaged.

☐
READ

☐
DREW
IT

☐
FROM
MEMORY